VirPhyKit

An integrated toolkit for viral phylogeographic analysis

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Contents

1. Introduction

VirPhyKit is a user-friendly stand-alone GUI-based software written in Python 3.10 and PyQt5. It functions allow users to:

- Retrieve, extract, and download sequences from GenBank, rename and group sequences, and plot sampling time and spatial information
- Subsample sequences, detect sampling bias, and reconstruct virus spatiotemporal migration patterns
- Visualize the posterior probability of the root state of the MCC tree and the Bayesian skyline.
- Perform molecular dating using TreeTime and TreeDater.

2. License & Disclaimer

2.1 License

This project is licensed under the GNU General Public License v3.0 - see the LICENSE file for details.

2.2 Disclaimer

This program is provided without any warranty. No warranty is given as to the functionality of the software, nor as to the accuracy of its results, either express or implied. Please check your results carefully.

3. Instructions

3.1 Installation

Installers for all platforms can be downloaded from https://github.com/BioEasy/VirPhykit

3.1.1 For Windows

Windows 7, 8 and 10 are all supported. Just download VirPhyKit.zip, unzip it, and then double-click VirPhyKit.exe to run it.

3.1.2 For Linux

Unzip or extract VirPhyKit.zip/VirPhyKit.tar.gz to your preferred directory, then double-click VirPhyKit.sh to run it. Or use the following commands:

```
cd path/to/VirPhyKit
./VirPhyKit.sh
```

If you encounter any permission-related errors, please try using the following command:

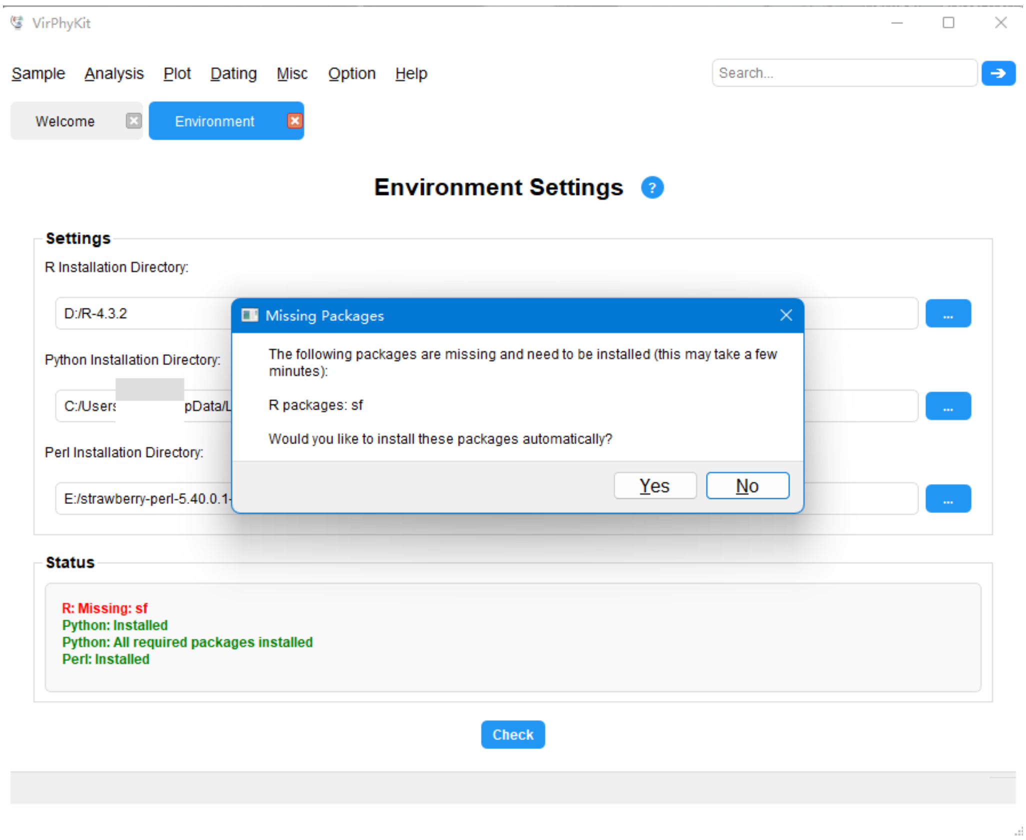
```
sudo chmod +x VirPhyKit.sh.
```

This will grant the VirPhyKit script execution permissions.

Note: Both 32-bit and 64-bit Windows (7 and above) are supported, but only 64-bit versions were tested on Linux (Ubuntu).

3.2 Environment check

Before using VirPhyKit, you need to specify the installation paths for R, Python, and Perl in 'Option-Environment' menu and click the [check] button. The system will automatically check if there are any missing Python packages and R packages. If so, the system will pop up a prompt indicating the missing package and provide an option to automatically download them.(On Linux, missing packages must be installed manually.)



To ensure the normal operation of each function, you need to manually or automatically download the following R and Python packages:

R	Python
tidyr	*numpy*
ggplot2	*pandas*
scales	*biopython*
ggsci	*ete3*
patchwork	
treeio	
plyr	
dplyr	
readr	
rnaturalearth	
rnaturalearthdata	
treedater	
ape	
maps	
sf	

4. Functions

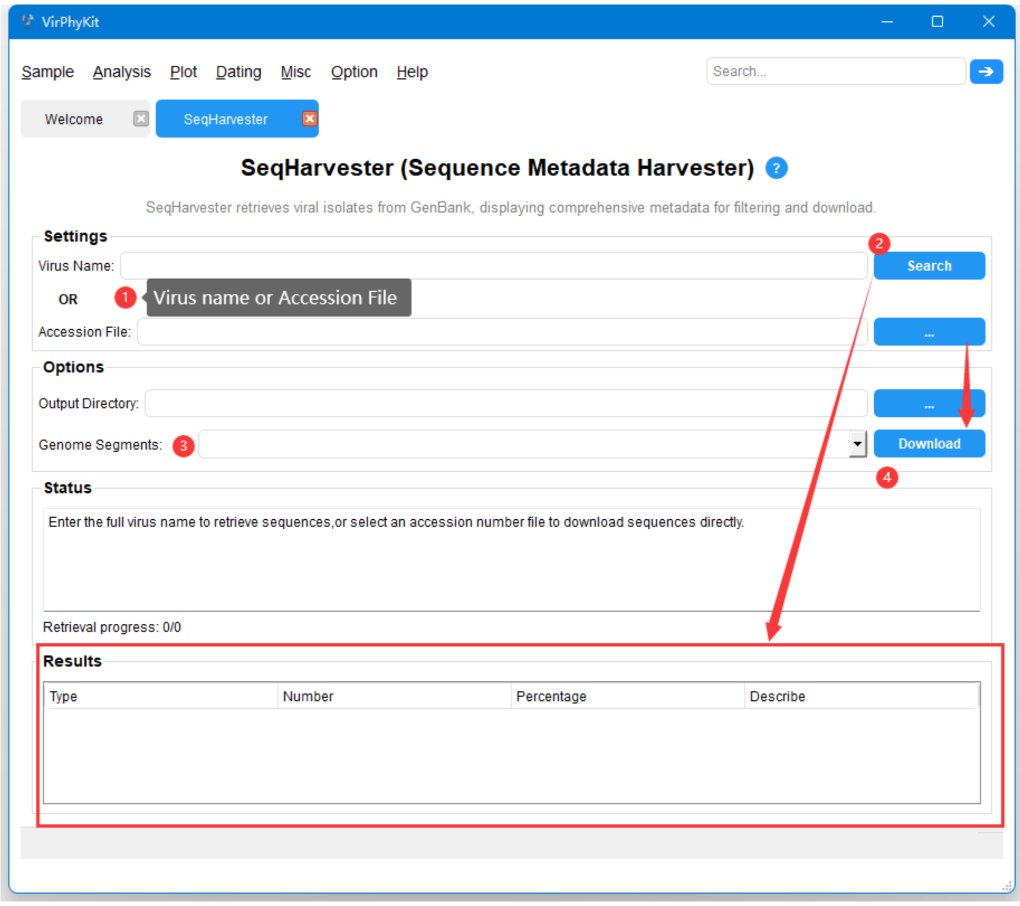
4.1 SeqHarvester

SeqHarvester retrieves viral isolates from GenBank, displaying comprehensive metadata for filtering and download.

4.1.1 Data Preparation

A txt file containing all the accession numbers of the target sequences (one per line)

4.1.2 Detailed Steps



Step 1: Enter the full name of the virus for retrieval or upload a text file containing the accession numbers.

Step 2: When retrieving by virus name, the results will display the complete metadata for all available isolates from GenBank in the ‘Results’ panel.

Step 3: From the ‘Genome Segments’ dropdown menu, select the required segment.

Step 4: Select the output directory, then click [Download] to initiate sequence retrieval.

Note: If using an accession number file, upload the file and specify the output directory, then simply click [Download] to start the download.

4.2 SeqIDRenamer

SeqIDRenamer is a tool for rapidly renaming sequence identifiers in FASTA files.This utility provides a simple workflow for batch processing of sequences IDs using custom naming rules.

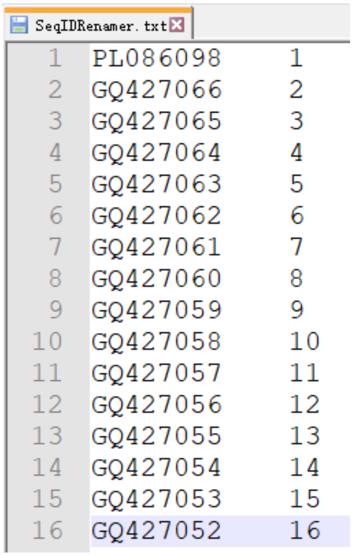
4.2.1 Data Preparation

If ‘SeqIDRenamer’ is used, a name mapping file (.txt) needs to be prepared. Each row of this file should contain the original name of the sequence and the new name you want to change it to, separated by a tab character. The format is as follows:

Orig1\tNew1

Orig2\tNew2

Orig3\tNew3

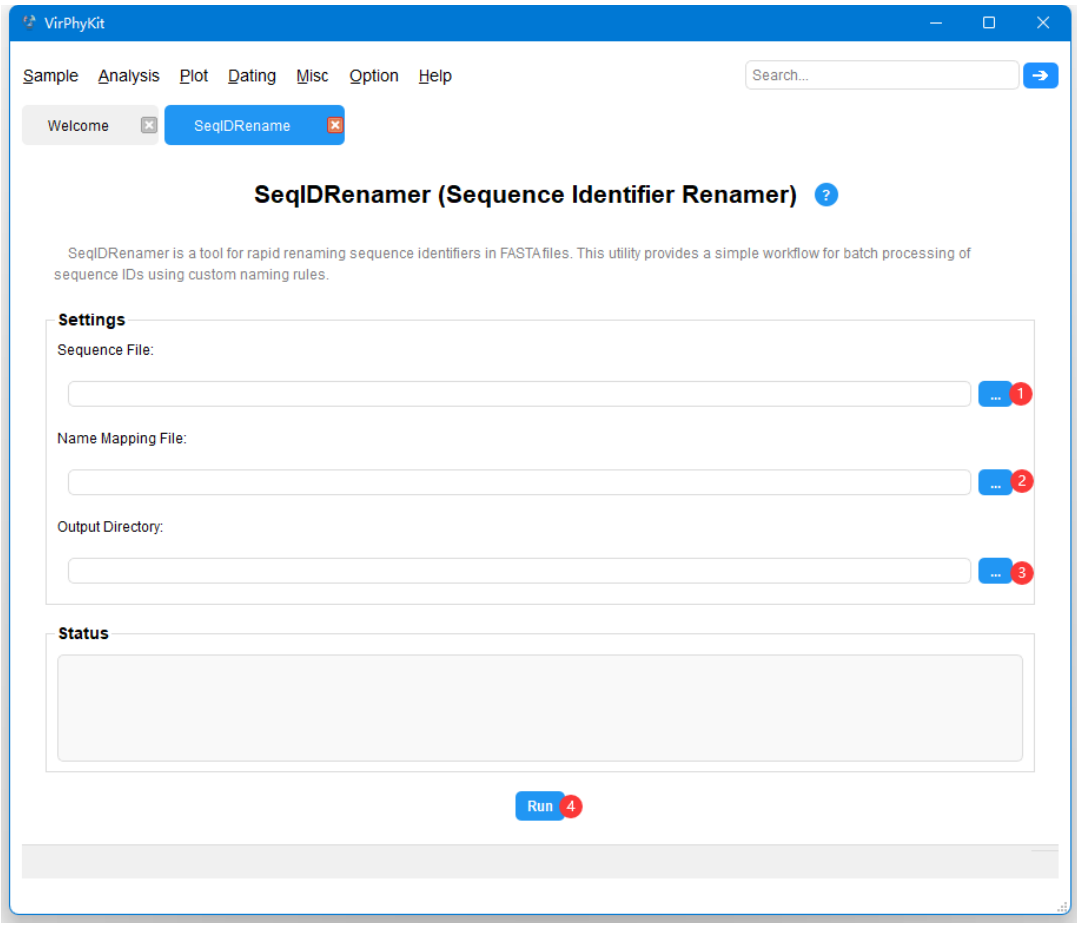


SeqIDRenamer.txt

PL086098	1
GQ427066	2
GQ427065	3
GQ427064	4
GQ427063	5
GQ427062	6
GQ427061	7
GQ427060	8
GQ427059	9
GQ427058	10
GQ427057	11
GQ427056	12
GQ427055	13
GQ427054	14
GQ427053	15
GQ427052	16

Name mapping file (.txt)

4.2.2 Detailed Steps



Step 1: Select a sequence file to rename.

Step 2: Upload a tab-delimited mapping text file (‘OrigName\tNewName’).

Step 3: Select the output directory to save the renamed file.

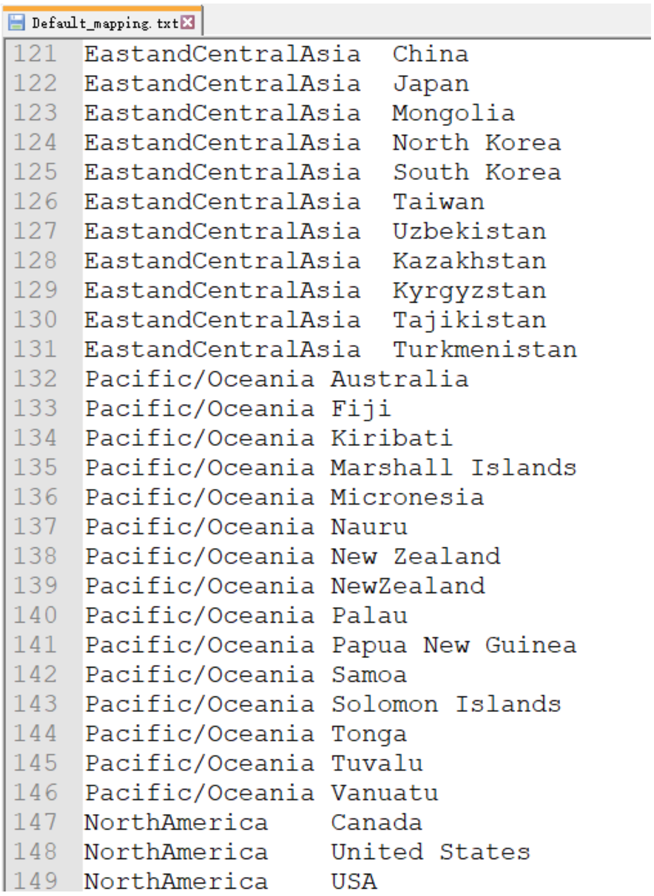
Step 4: Click the [Run] button to start the renaming process.

4.3 SeqGrouper

SeqGrouper is a specialized tool for grouping viral sequences into customizable categories(e.g., geographic region, host, or user-defined criteria), with automated distribution reports for research and surveillance.

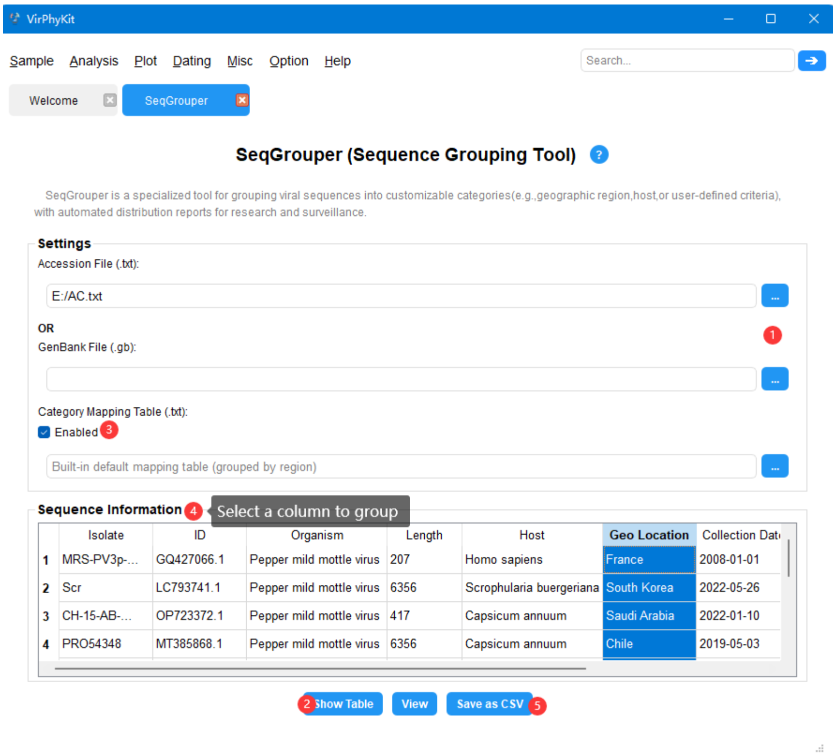
4.3.1 Data Preparation

A grouped mapping table file (.txt) is provided for mapping the attributes (such as geographical location or host) in the sequence data to custom groups (e.g., mapping "China" to "East and Central Asia"). Each line of the file contains two fields, separated by a tab: the first field is the group name, and the second field is the field value. You can either use the default file or load a custom file to generate a distribution pie chart. Make sure to use UTF-8 encoding and the correct format.



```
Default_mapping.txt
EastandCentralAsia	China
EastandCentralAsia	Japan
EastandCentralAsia	Mongolia
EastandCentralAsia	North Korea
EastandCentralAsia	South Korea
EastandCentralAsia	Taiwan
EastandCentralAsia	Uzbekistan
EastandCentralAsia	Kazakhstan
EastandCentralAsia	Kyrgyzstan
EastandCentralAsia	Tajikistan
EastandCentralAsia	Turkmenistan
Pacific/Oceania	Australia
Pacific/Oceania	Fiji
Pacific/Oceania	Kiribati
Pacific/Oceania	Marshall Islands
Pacific/Oceania	Micronesia
Pacific/Oceania	Nauru
Pacific/Oceania	New Zealand
Pacific/Oceania	NewZealand
Pacific/Oceania	Palau
Pacific/Oceania	Papua New Guinea
Pacific/Oceania	Samoa
Pacific/Oceania	Solomon Islands
Pacific/Oceania	Tonga
Pacific/Oceania	Tuvalu
Pacific/Oceania	Vanuatu
NorthAmerica	Canada
NorthAmerica	United States
NorthAmerica	USA
```

4.3.2 Detailed Steps



Step 1: Upload accession numbers in a text file (.txt) or upload a GenBank file (.gb).

Step 2: After checking 'Enable' checkbox, you can choose to use the default grouping rules (by region) or upload a custom grouping table.

Step 3: Click [Show Table] to display viral isolate information (You can double-click any information to modify it).

Step 4: Select the columns that need to be grouped and click [View] to display the group distribution.

Note: The grouped columns will be added to the table.

4.4 VirSpaceTime

VirSpaceTime is a tool for visualizing spatial or temporal distribution patterns of viral isolates.

4.4.1 Input and Output

Input:

Temporal file (.txt): Contains the sample quantity of each region for each year, in tab-delimited format (see Example 1).

Spatial file (.txt): Contains the latitude and longitude coordinates of the sampling locations, in tab-delimited format (see Example 2).

Output: Spatiotemporal distribution map in PDF format.

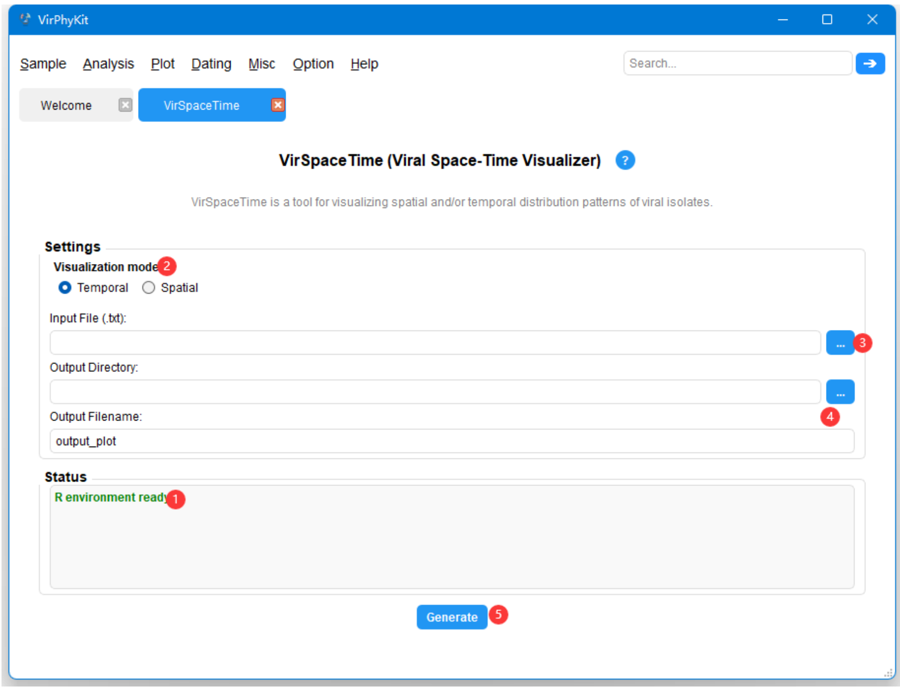
Year	EA	EU	ME	AM	SA	MC	Total
1997	0	1	0	0	0	0	1
1998	0	0	0	0	0	1	1
2003	0	1	1	0	0	0	2
2005	0	1	0	1	0	0	2
2006	0	1	0	1	0	0	2
2008	2	1	0	0	0	0	3
2010	0	1	0	4	0	0	5
2011	0	7	0	0	0	1	8
2012	1	0	12	0	0	7	20
2013	11	0	0	0	0	0	11
2014	0	1	0	0	0	17	18
2015	0	0	7	1	0	12	20
2016	0	1	0	0	1	5	7
2017	3	0	0	0	0	8	11
2018	1	0	5	0	5	0	11
2019	0	0	0	3	2	17	22
2020	1	0	0	1	0	0	2
2021	1	4	0	0	0	0	5
2022	0	0	2	0	0	1	3
2023	1	0	0	0	0	0	1

Example1: Temporal file(.txt)

region	Latitude	Longitude
EA	36.5625	139.8836
EA	37.5665	126.9780
EA	25.0320	121.5654
EA	32.7897	130.7417
EA	33.6823	130.7017
EA	35.6823	139.7662
MC	36.6758	117.0009
MC	31.9686	118.6946
MC	38.0438	114.5149
MC	39.9042	116.4074
MC	34.4205	113.6465
MC	37.3743	118.3124
MC	25.0463	102.7103
MC	41.4300	123.1925
MC	34.7756	108.1056
MC	31.9735	112.2981
MC	36.7030	116.3420
MC	32.0493	118.7893
MC	35.1121	112.2785
MC	37.6543	114.0361
MC	34.7200	113.7101
MC	32.2126	120.6750
MC	33.5490	118.9735
ME	31.2357	30.0444
AM	38.9072	-77.0369
EU	48.8566	2.3522

Example 2:Spatial file (.txt)

4.4.4 Detailed Steps



Step 1: Ensure R is properly configured (in the ‘Option-Environment’ menu).

Step 2: Select visualization mode (spatial distribution or temporal distribution).

Step 3: Upload a tab-delimited text file, containing geographic coordinates for spatial analysis or population sample sizes over time for temporal analysis.

Step 4:Select the output directory and filename.

Step 5: Click [Generate] to create the visualization.

Notes:

- Make sure the input file is encoded in UTF-8 and fields are separated by tabs (\t).
- Check if the R dependencies are installed .
- If the generated result is empty, please confirm whether the input file format is correct.

4.5 RRT

RRT is a statistical method that detects sampling bias in phylogeographic analyses. It compares ancestral root state probabilities between the original dataset and location-permuted scenarios to correct for systematic overestimation of viral origins in oversampled regions.

4.5.1 Input and Output

Input:

Original MCC tree: An annotated MCC tree generated by BEAST 1.x or BEAST 2.x.

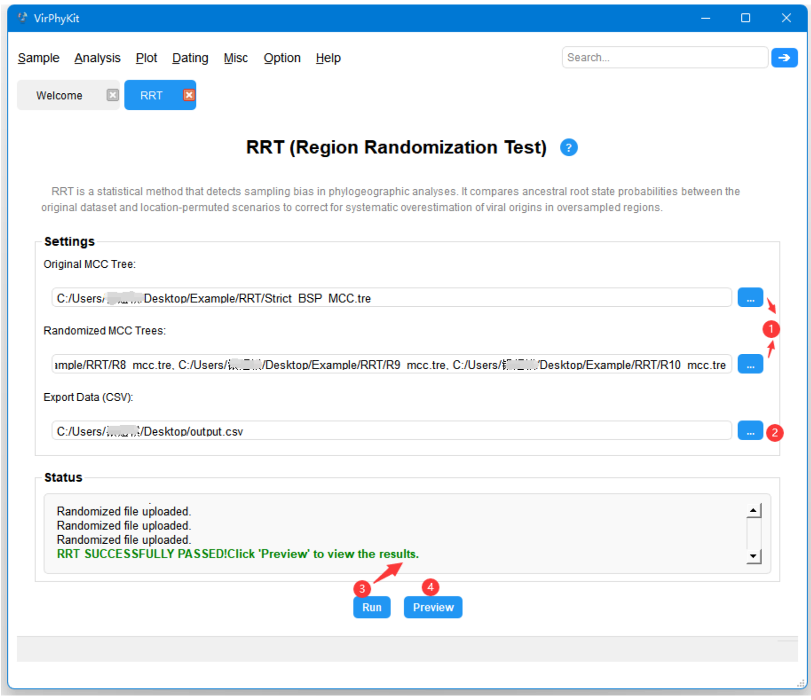
Randomized MCC tree: Multiple region-randomized MCC tree.

Output:

CSV table: Comparison results of posterior probabilities of ancestral root states for the original and randomized datasets.

Status panel: Displays the RRT test results, indicating whether there is a significant sampling bias.

4.5.2 Detailed Steps



Step 1:Upload the original MCC tree and the region-randomized MCC tree.

Step 2:Select an output directory for the posterior probability CSV table.

Step 3:Click [Run] to conduct the regional randomization tests. The test results will be displayed on the Status panel.

Step 4:Specify the output directory for saving the results.

4.6 TempMig/TempMig Plotter

TempMig is a specialized tool for reconstructing and visualizing temporal migration patterns of viral pathogens. The utility processes an MCC tree annotated with traits or a MultiTypeTree to generate time-resolved migration matrices and interactive visualizations.

4.6.1 Data Preparation

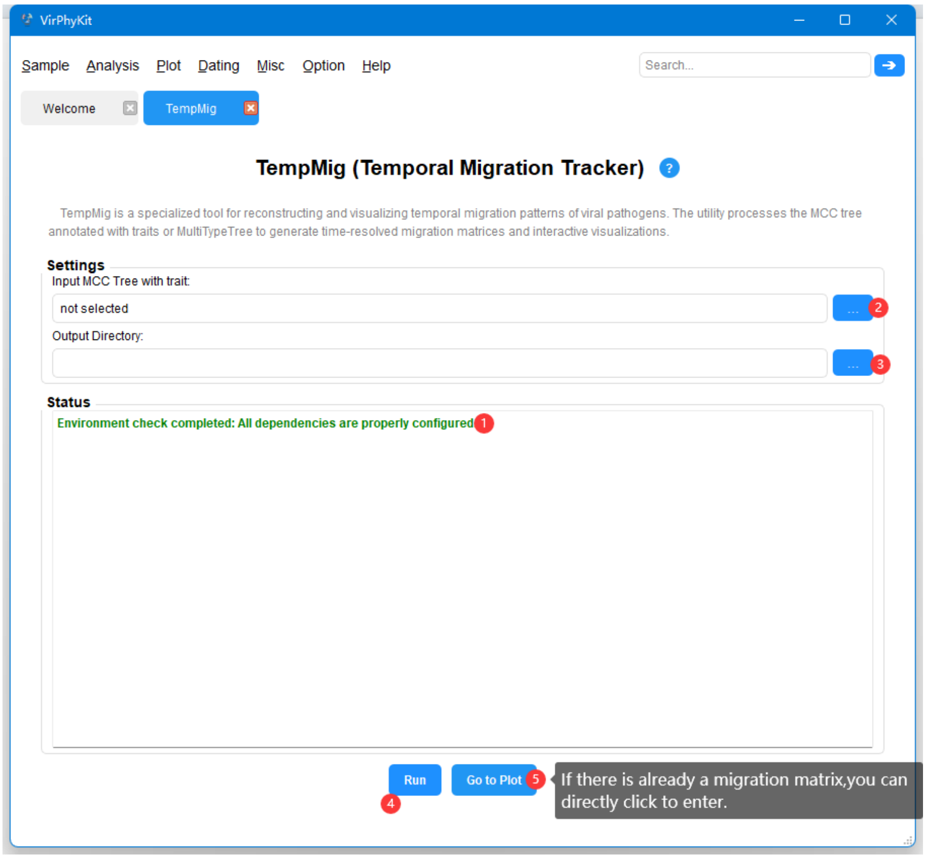
An MCC tree or MultiTypeTree with trait annotations.

4.6.2 Detailed Steps

TempMig:

Input: An MCC Tree annotated with traits or a MultiTypeTree.

Output: Migration matrix file (.txt) for TempMig Plotter.



Step 1: Ensure the Python, Perl and R installation directories are configured in the ‘Option-Environment’ menu.

Step 2: Upload an MCC tree annotated with traits or a MultyTypeTree.

Step 3: Specify the output directory for the migration matrix.

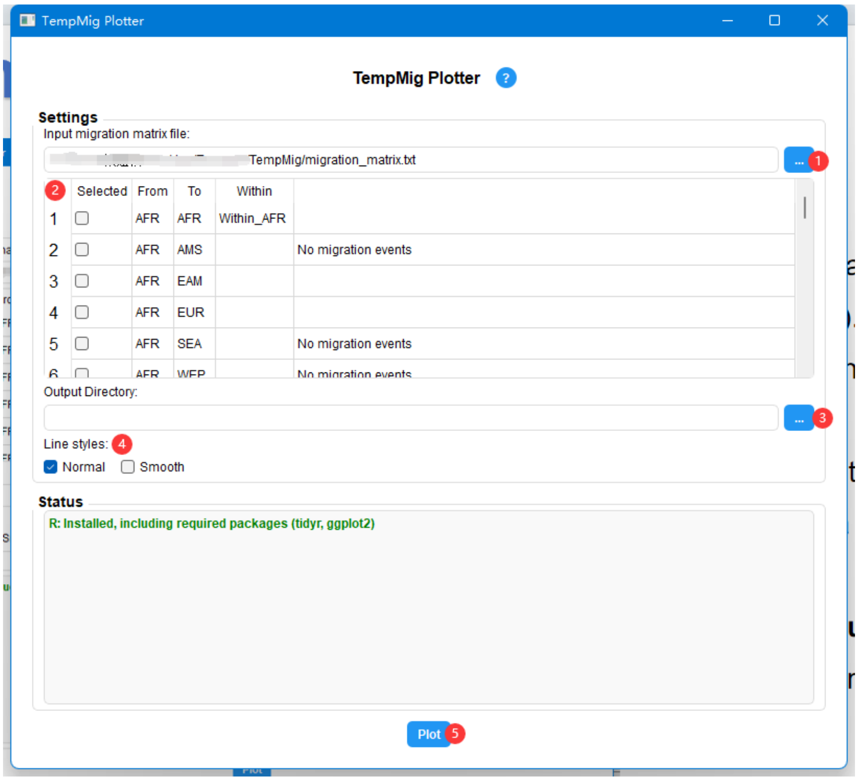
Step 4: Click [Run] to process the MCC tree/MultiTypeTree and generate the migration matrix.

Step 5: Click [Go to plot] to move to the ’TempMig Plotter’ tool and visualize the results using the migration matrix.

TempMig Plotter:

Input: Migration matrix file (.txt) from ‘TempMig’

Output: Temporal migration patterns(.PDF)



Step 1: Upload a migration matrix file (output from ’TempMig’).

Step 2: Select the migration directions for visualization.

Step 3: Specify the output directory.

Step 4: Select a visualization style (Normal or Smooth).

Step 5: Click [Run] to generate the temporal migration patterns.

4.7 GeoSubsampler

GeoSubsampler is a specialized tool for region-stratified subsampling of FASTA-formatted sequence datasets. It generates balanced sequence subsets by either geographic region or sample size, with integrated support for bootstrap analysis.

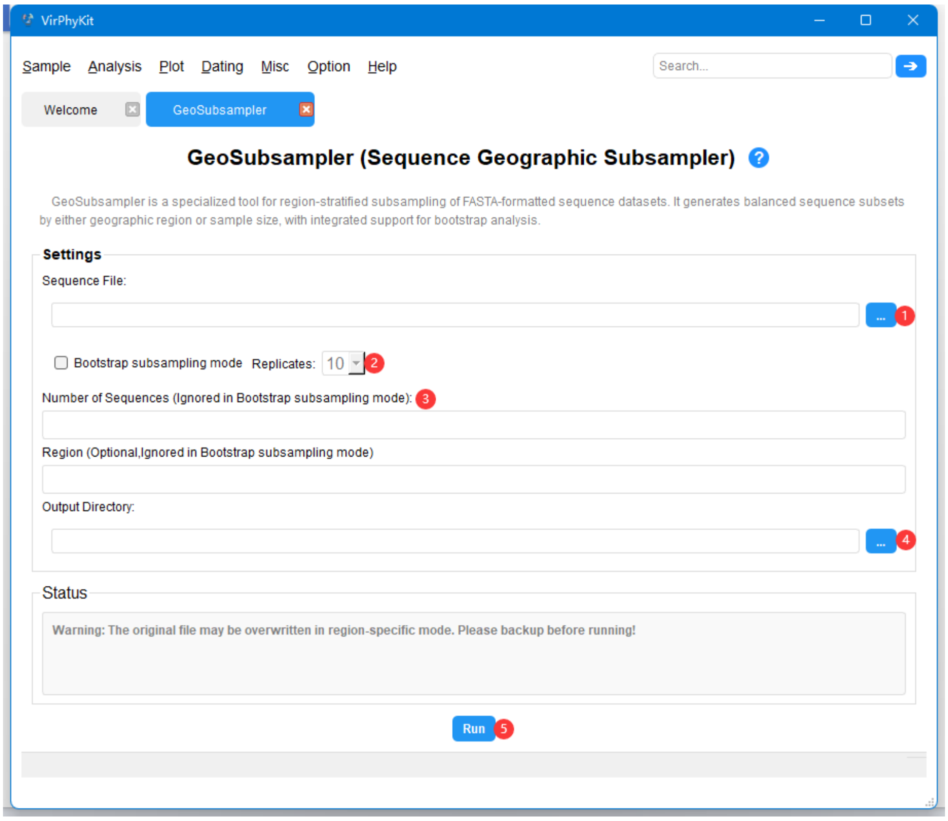
4.7.1 Data Preparation

FASTA sequence file. The naming of each sequence must contain region information (e.g., Region_, _Region_, _Region). This function will conduct sub-sampling from the original file and save it as a new file. Therefore, please make sure to back up the original file before using it.

4.7.2 Detailed Steps

Input: FASTA sequence file.

Output: The files that were sub-sampled according to the set parameters



Step 1: Upload a FASTA sequence file to be subsampled. Ensure each sequence name includes its geographic region (marked with '_region').

Step 2: (Optional) Enable the 'Bootstrap subsampling mode' option and set the number of replicates.

Step 3: Specify the desired sample size or specific region for the subset.

Step 4: Select an output directory to save the resulting sequences.

Note on Bootstrap subsampling mode: In this mode, users do not need to specify the number of subsamples or select specific regions. The system automatically identifies the region with the fewest data entries in the sequence file and uses this number as the standard to perform uniform subsampling across all regions.

4.8 RSPP-Viz

RSPP-Viz is a specialized tool for visualizing root state posterior probabilities inferred from MCC trees or MultiTypeTree.

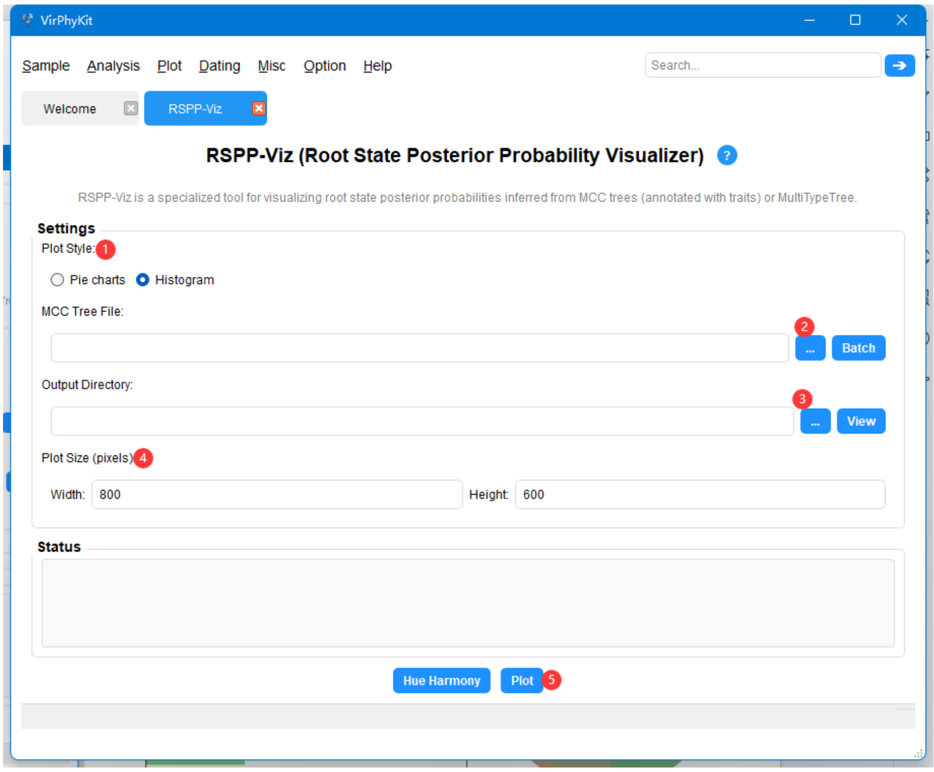
4.8.1 Data Preparation

MCC trees (annotated with traits) or MultiTypeTrees.

4.8.2 Detailed Steps

Input: MCC trees or MultyTypeTree.

Output: RSPP plot (.PDF)



Step 1: Select the plot style you want to generate (pie or histogram).

Step 2: Select the MCC tree (annotated with traits) or MultiTypeTree for analysis (batch processing is available in Batch mode).

Step 3: Specify the output directory.

Step 4: Specify the plot size.

Step 4: Use [Hue Harmony] to adjust color schemes, and click [Plot] to create the plot.

4.9 BSP-Viz

BSP-Viz is a specialized tool for visualizing Bayesian Skyline Plots(BSP), supporting both single-file and batch processing modes.

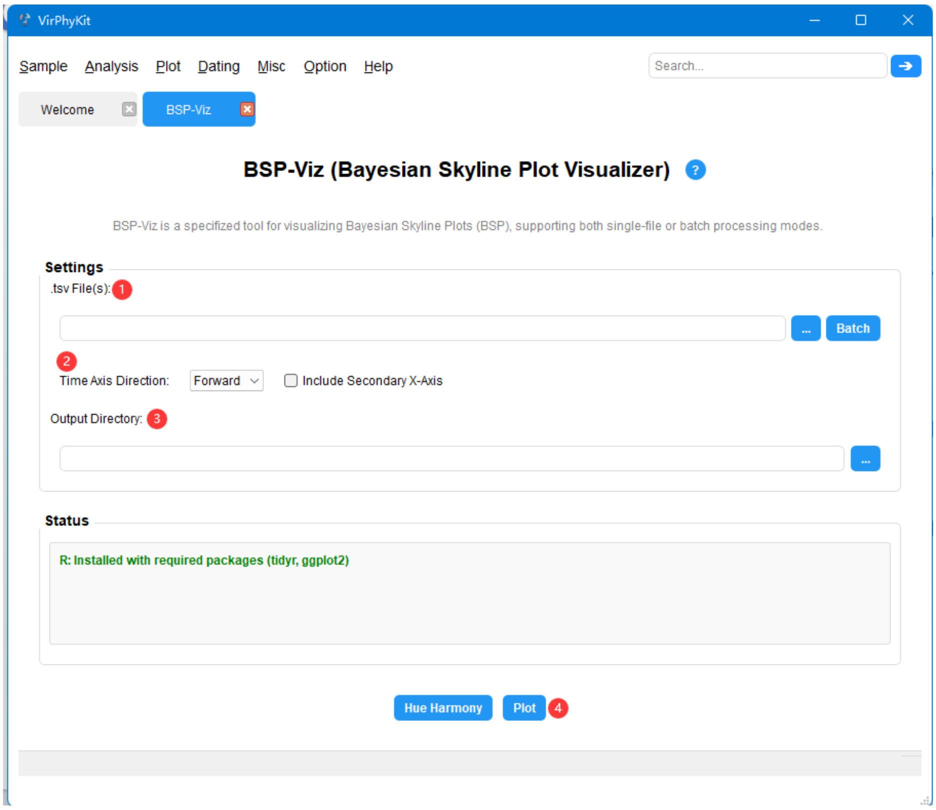
4.9.1 Data Preparation

To visualize a Bayesian skyline plot, import the log files from a BEAST Bayesian skyline analysis into Tracer, select ‘Analysis-Bayesian Skyline Reconstruction’ and view the population dynamics graph. Export the skyline data directly as a .tsv file.

4.9.2 Detailed Steps

Input: Bayesian Skyline data file (.tsv)

Output: One or more (in batch mode) BSP plots.



Step 1: After ensuring that the R environment is configured correctly, select the table file (you can click the [Batch] button to draw in batches later).

Step 2: Select the time axis direction and whether to add a secondary axis.

Step 3: Specify the output directory and click [Hue Harmony] to adjust color schemes.

Step 4: Click [Plot] to draw.

4.10 TreeTime-RTT

TreeTime-RTT is a tool for assessing temporal signal (lock-like evolutionary patterns) through linear regression of root-to-tip distance against sampling dates in TreeTime.

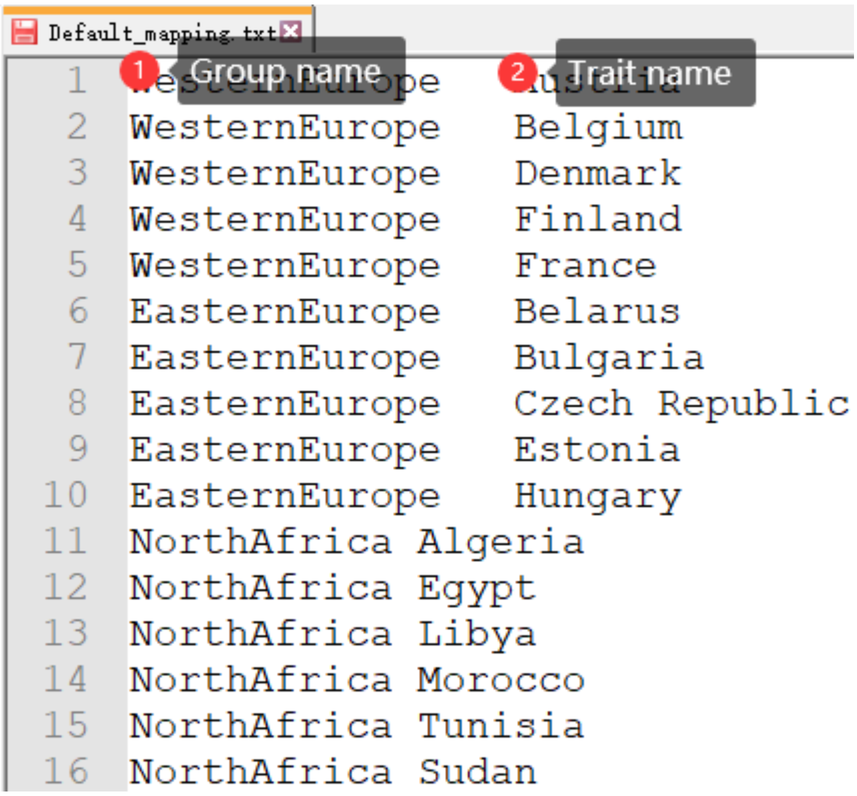
4.10.1 Data Preparation

a. To use ‘TreeTime-RTT’, prepare a metadata file(.csv) with the first column listing Newick tree tip names, the second column specifying corresponding sampling times, and an optional third column for trait names to group data points in the RTT linear regression scatter plot.(Taking the location trait as an example)

	1 tip name	2 sampling date	3 trait name(optional)
1	name	date	location
2	A/HaNoi/Q545/2007|C	2007.569473	VietNam
3	A/Texas/UR06_0358/2	2007.13963	USA
4	A/Hong_Kong/CUHK528	2004.629706	HongKong
5	A/Peru/PER004/2010|	2010.887064	Peru
6	A/Waikato/1/2000|CY	2000.607803	NewZealand
7	A/Hong_Kong/1179/99	1999.002738	HongKong
8	A/West_Virginia/17/	2012.596851	USA
9	A/New_York/913/2005	2005.027379	USA
10	A/Novosibirsk/59d/2	2012.202601	Russia

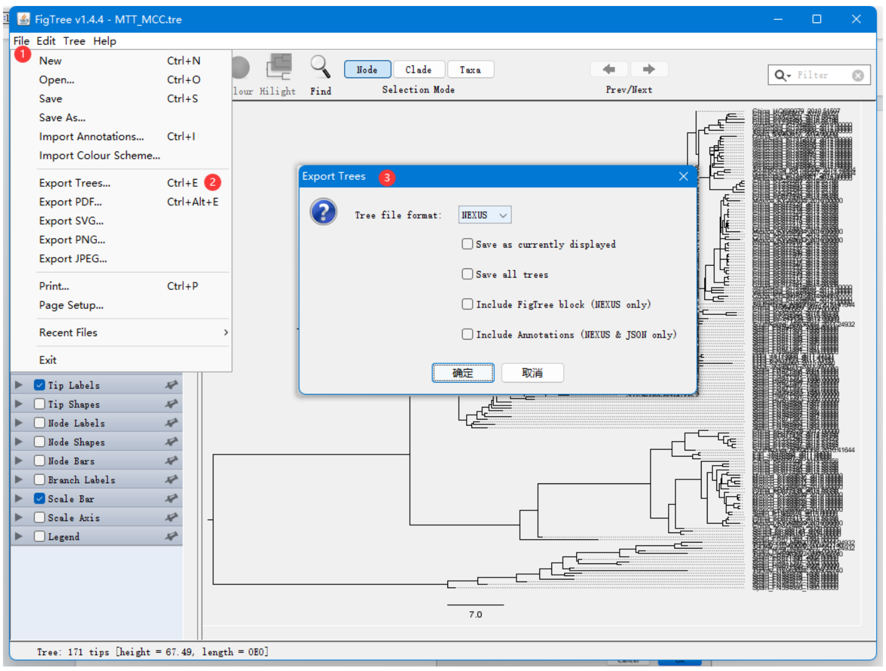
Metadata file(.csv)

b. If you want to group the scattered points in the RTT linear regression plot, you need to prepare a trait name grouping mapping file (.txt) at the same time (taking location trait as an example)



Mapping file(.txt)

c. The tree file must be in Newick format. The branch length can be included, and each node can have a label. The names of classification units cannot contain illegal characters (such as spaces and semicolons).

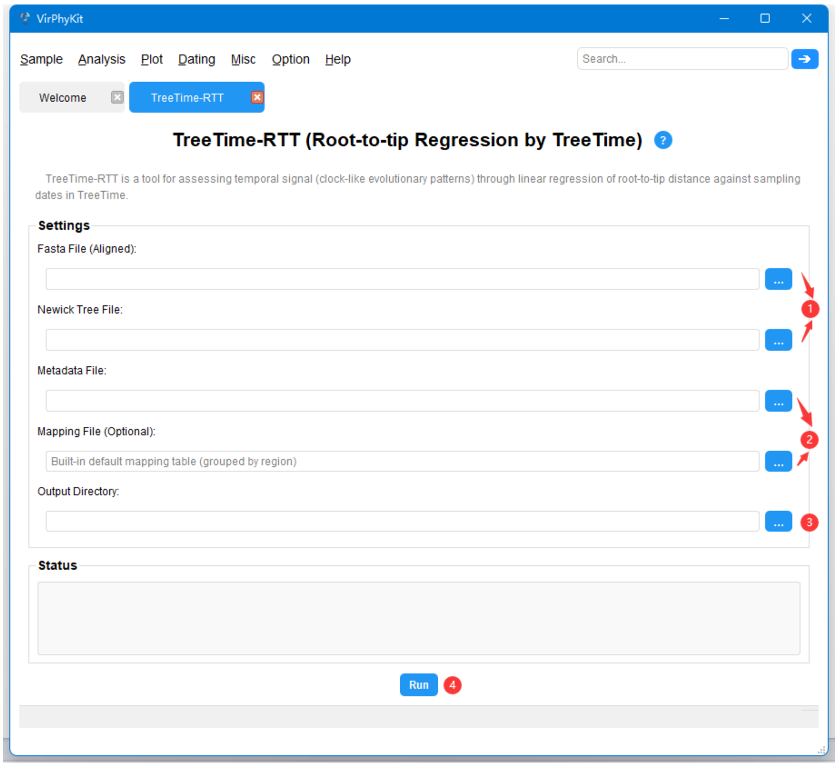


Newick Tree(.nwk)

4.10.2 Detailed Steps

Input: FASTA file(aligned), Newick tree, metadata file,and mapping file.

Output: Root-to-tips linear regression plot,and Time-Inference tree(.nwk,.pdf)



Step 1: Upload a FASTA sequence file, a Newick tree file, and a metadata file (.csv).

Step 2: If the third column of the metadata file is a “region”(or other trait) column and a trait mapping file is uploaded, isolates sharing the same trait will be color-coded in the plot. Otherwise, only the standard RTT regression plot will be generated.

Step 3: Select the output directory.

Step 4: Click [Run] to perform the RTT analysis.

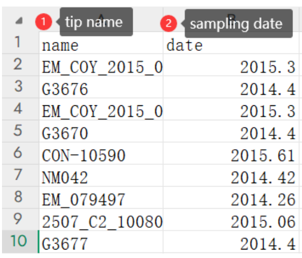
4.11 TreeDater-LTT

TreeDater-LTT is a tool for conducting a lineage-through-time (LTT) analysis using Treedater. Additionally, it can also infer the time to the most recent ancestor (TMRCA) and the substitution rate, implemented with a scalable relaxed-clock phylogenetic dating framework.

4.11.1 Data Preparation

a. Newick tree(.nwk)

b. The metadata file(.txt) has the first column as "tip name" and the second column as "sampling date".



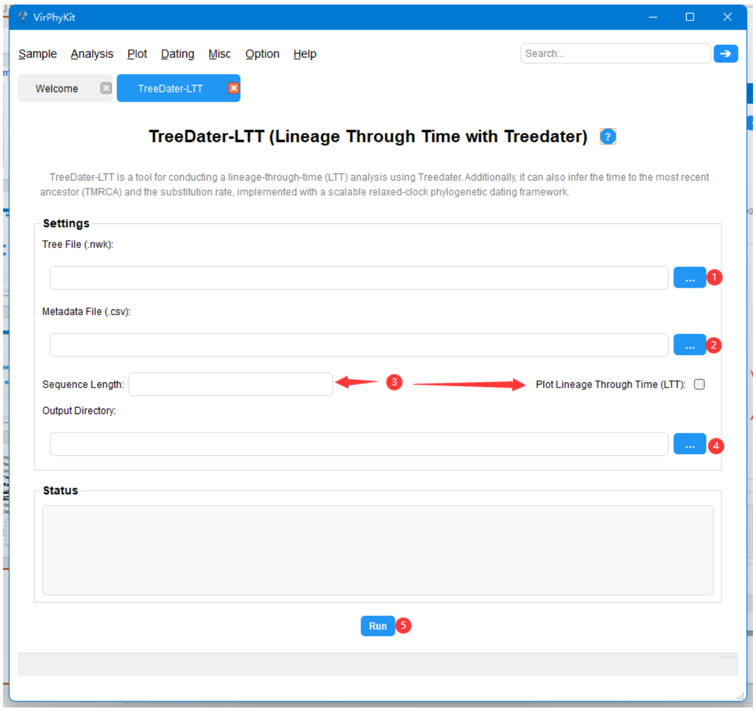
1	name	date
2	EM_COY_2015_0	2015.3
3	G3676	2014.4
4	EM_COY_2015_0	2015.3
5	G3670	2014.4
6	CON-10590	2015.61
7	NM042	2014.42
8	EM_079497	2014.26
9	2507_C2_10080	2015.06
10	G3677	2014.4

Metadata file(.txt)

4.11.2 Detailed Steps

Input: Newick tree (.nwk), Metadata file (.txt)

Output: The results related to TMRCA displayed in the Status panel, and a lineage-through-time plot (LTT)



Step 1: Select a Newick tree file (.nwk).

Step 2: Upload the metadata file (.csv) with sampling dates.

Step 3: Enter the sequence length and enable the 'LTT' option to generate a lineage-through-time plot.

Step 4: Specify the output directory for saving results.

Step 5: Click [Run] to perform the LTT analysis using TreeDater, and view the results in the Status panel and output images.

4.12 MJRM Generator

MJRM is a specialized tool that automatically constructs Markov jump and reward matrices from user-defined discrete traits for phylogeographic analysis. It seamlessly integrates these matrices into BEAST-compatible XML configuration files.

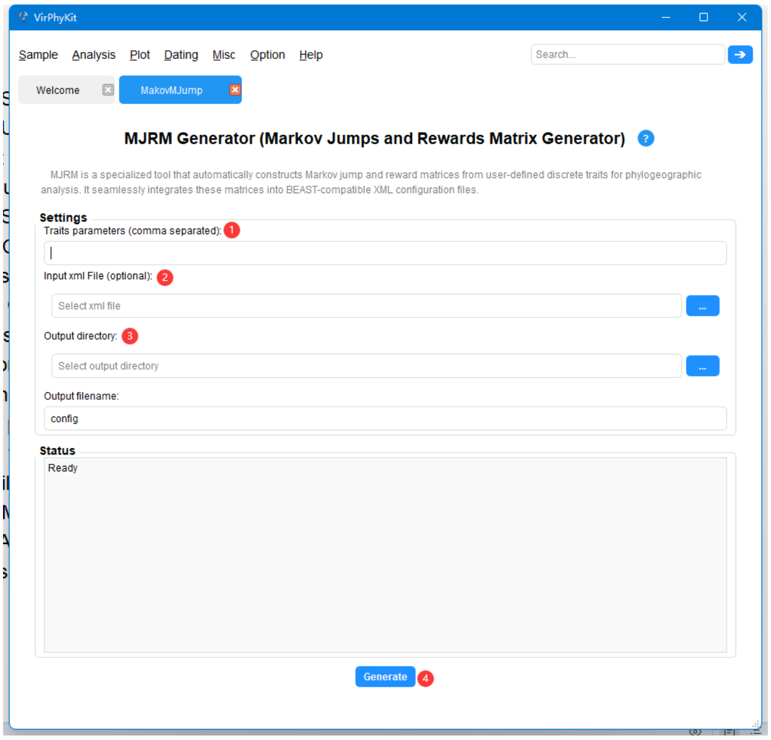
4.12.1 Data Preparation

An XML file configured in BEAUti which contains the defined traits

4.12.2 Detailed Steps

Input: XML configured file.

Output: A new XML configuration file that automatically incorporates Markov jumps and reward matrices and can be directly used for BEAST analysis.



Step 1: Specify the discrete trait categories (e.g. location, host, etc.) as comma-separated values.

Step 2: Upload a BEAUti 1.x-generated xml file for modification.

Step 3: Specify the output directory and name for the new xml file.

Step 4: Click [Run] to automatically generate and insert the matrix module into the new xml file.

Note: If you choose not to upload the XML file, the generated Markov jump and reward matrix will be saved in txt format in the specified output directory.

5. VirPhyKit Slow Startup

When starting VirPhyKit for the first time after the device is powered on, a longer waiting period (usually ten seconds) is required. This is because the system needs additional time to initialize the relevant resources.

6. Citation

If you use data generated by VirPhyKit in a scientific paper, please use the following citation:

Yin et al., 2025. VirPhyKit: an integrated toolkit for viral phylogeographic analysis.

7. Acknowledgements

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